



Advancing personalized risk assessment in post-TIPS hepatic encephalopathy: insights and future directions

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Received: 28 November 2025 / Accepted: 11 January 2026
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We read with great interest the article by Miao et al. using TabPFN-based machine learning to predict hepatic encephalopathy (HE) after transjugular intrahepatic portosystemic shunt (TIPS) placement [1]. The authors reported strong predictive performance, with AUCs of 0.953, 0.870, and 0.942 across training, validation, and external test sets, outperforming traditional clinical scores such as MELD (AUC 0.573 in external testing). We commend this integrative approach combining radiomics and deep learning features from multiple organs, addressing an important clinical need in portal hypertension management.

We would like to offer some friendly observations that might strengthen future work in this area. The study's 20.6% overall HE incidence appears notably lower than the typically reported 20–50% range [2], which may reflect the specific patient selection or follow-up protocols used. It would be valuable to understand whether baseline characteristics like Child–Pugh distribution or TIPS indication patterns (86.6% variceal bleeding in training set versus 77.1% in external set) contributed to this difference, as such variations could influence model generalizability across different clinical settings. Understanding these demographic nuances could help clinicians better interpret risk predictions when applying this model to their own patient populations.

Although liver radiomics features emerged as the dominant predictors in SHAP analyses, the routine clinical implementation of automated liver segmentation using TotalSegmentator warrants additional consideration. Prior studies have shown that segmentation performance may vary with

imaging quality and cirrhosis-related morphological heterogeneity [3]. Further validation in cirrhotic livers with nodular contours, regional atrophy, or ascites, as well as reporting the frequency of manual corrections, would enhance confidence in real-world deployment.

The exclusion of patients lost to follow-up within 12 months is reasonable, given that most post-TIPS HE events occur during this period [4]. Nonetheless, supplementary sensitivity analyses could help mitigate concerns regarding potential selection bias. In addition, while serum ammonia levels were not included, their integration in future iterations may enhance both predictive accuracy and clinical interpretability, given their central role in HE pathophysiology [5].

Looking ahead, further refinement of HE risk prediction may benefit from broader consideration of data modalities beyond the current study's scope. More advanced imaging approaches—such as multiphase CT radiomics, functional MRI techniques, or quantitative perfusion imaging—may capture subtle hemodynamic and metabolic alterations preceding overt HE. Incorporating novel clinical features, including frailty indices, neurocognitive screening measures, and medication burden, could further enrich risk stratification. Moreover, integrating multi-omics data, such as metabolomic profiles, gut microbiome signatures, or inflammatory biomarkers, with imaging-derived features represents a promising avenue toward a more mechanistically informed, precision-oriented framework for post-TIPS HE risk assessment.

The time-dependent ROC analysis showing AUCs exceeding 0.88 at 90, 180, 365, and 730 days is particularly valuable for clinical decision-making, as it allows risk assessment at multiple clinically relevant time points. This temporal dimension could guide the timing and intensity of preventive interventions in high-risk patients identified by the model.

We believe this work represents an important step toward precision medicine in TIPS management and look forward to

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seeing prospective validation studies that incorporate these considerations.

Author contributions WC: formal analysis, writing-original draft, writing-review & editing. ZY: formal analysis, investigation, writing-review & editing. The author has read and agreed to the published version of the manuscript.

Funding The authors declare that no external funding was received for this work.

Data availability Data sharing is not applicable to this article as no new data were created or analyzed in this study.

Declarations

Conflict of interest The authors declare no competing interests.

Research involving human participants and/or animals This research did not require the involvement of human or animal subjects. Therefore, ethical approval for this study was not required according to local regulations and institutional policies.

Informed consent Not applicable.

Use of generative AI During the preparation of this work, the authors used Claude Opus 4.5 and ChatGPT-5 solely to improve the readability and language quality of the manuscript. The AI tools were used exclusively for linguistic refinement and did not generate any new intellectual content or original data. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the integrity and content of the published article.

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